

Iterated integrals

PARTIAL INTEGRATION

Suppose that f is a function of two variables that is integrable on the rectangle $R = [a, b] \times [c, d]$. We use the notation $\int_c^d f(x, y) dy$ to mean that x is held fixed and $f(x, y)$ is integrated with respect to y from $y = c$ to $y = d$. This procedure is called *partial integration with respect to y* . (Notice its similarity to partial differentiation.) Now $\int_c^d f(x, y) dy$ is a number that depends on the value of x , so it defines a function of x :

$$A(x) = \int_c^d f(x, y) dy$$

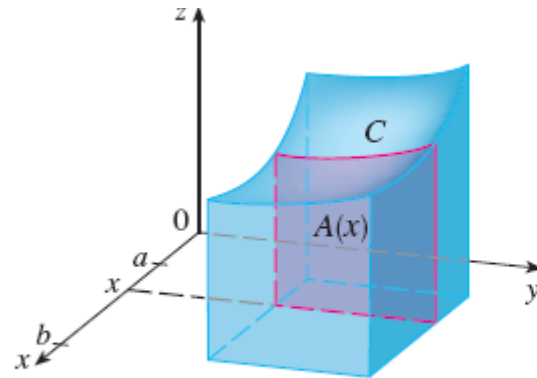


FIGURE 1

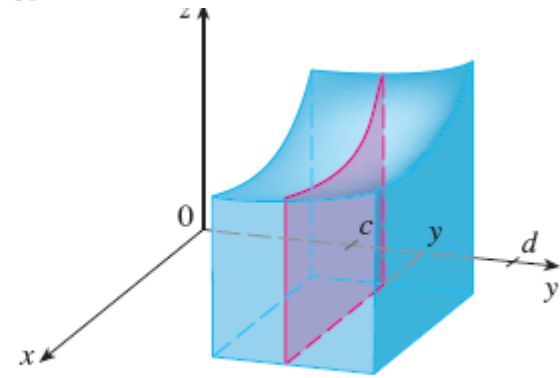


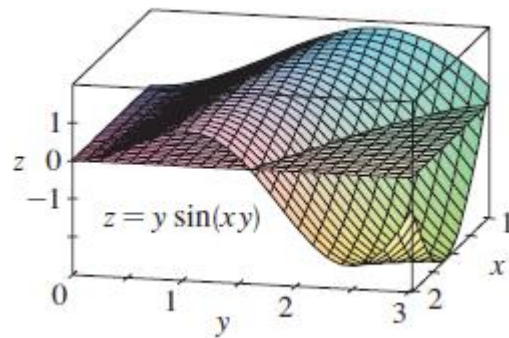
FIGURE 2

10 Fubini's Theorem If f is continuous on the rectangle $R = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$, then

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

More generally, this is true if we assume that f is bounded on R , f is discontinuous only on a finite number of smooth curves, and the iterated integrals exist.

EXAMPLE 6 Evaluate $\iint_R y \sin(xy) \, dA$, where $R = [1, 2] \times [0, \pi]$.



For a function f that takes on both positive and negative values, $\iint_R f(x, y) \, dA$ is a difference of volumes: $V_1 - V_2$, where V_1 is the volume above R and below the graph of f , and V_2 is the volume below R and above the graph. The fact that the integral in Example 6 is 0 means that these two volumes V_1 and V_2 are equal. (See Figure 14.)

In the special case where $f(x, y)$ can be factored as the product of a function of x only and a function of y only, the double integral of f can be written in a particularly simple form. To be specific, suppose that $f(x, y) = g(x)h(y)$ and $R = [a, b] \times [c, d]$. Then Fubini's Theorem gives

$$\boxed{11} \quad \iint_R g(x) h(y) \, dA = \int_a^b g(x) \, dx \int_c^d h(y) \, dy \quad \text{where } R = [a, b] \times [c, d]$$

Double integrals over general regions

1

$$F(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \text{ is in } D \\ 0 & \text{if } (x, y) \text{ is in } R \text{ but not in } D \end{cases}$$

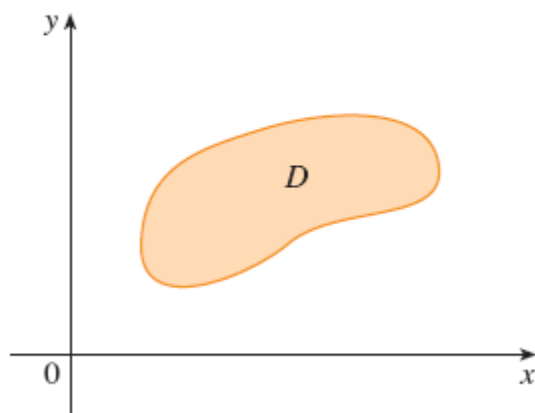


FIGURE 1

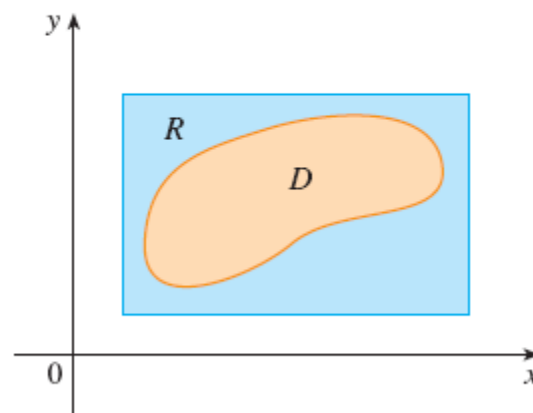


FIGURE 2

If F is integrable over R , then we define the **double integral of f over D** by

$$\boxed{2} \quad \iint_D f(x, y) \, dA = \iint_R F(x, y) \, dA \quad \text{where } F \text{ is given by Equation 1}$$

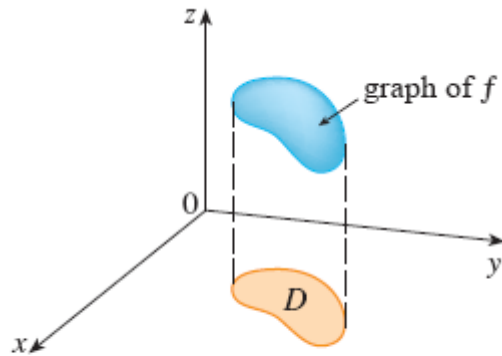


FIGURE 3

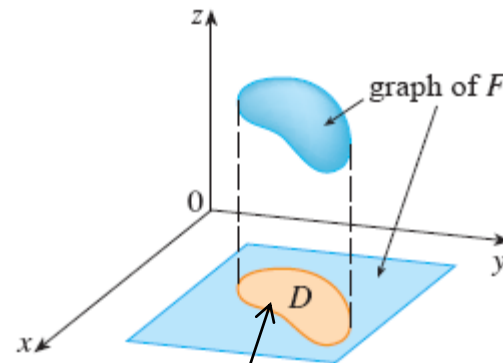


FIGURE 4

Any comment on the nature of $F(x,y)$ on the boundary of D ?

A plane region D is said to be of **type I** if it lies between the graphs of two continuous functions of x , that is,

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

where g_1 and g_2 are continuous on $[a, b]$. Some examples of type I regions are shown in Figure 5.

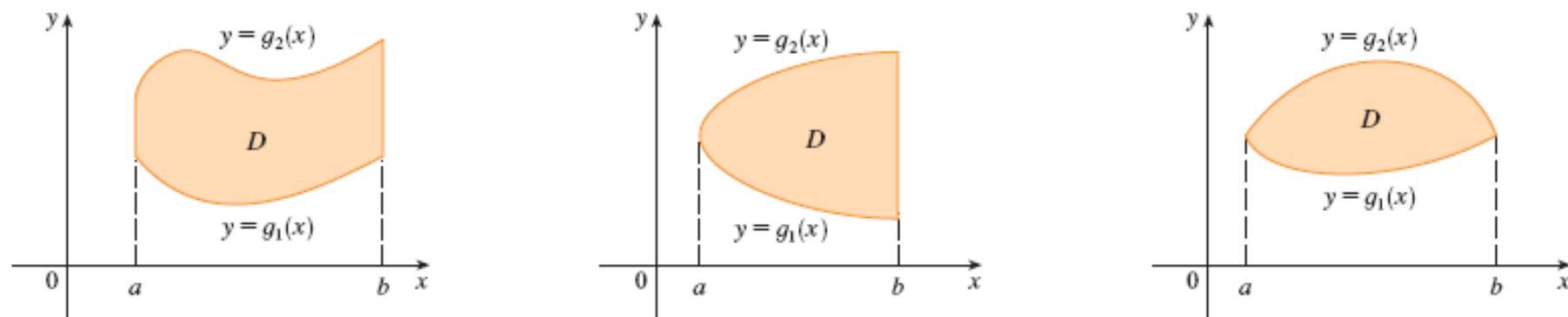


FIGURE 5 Some type I regions

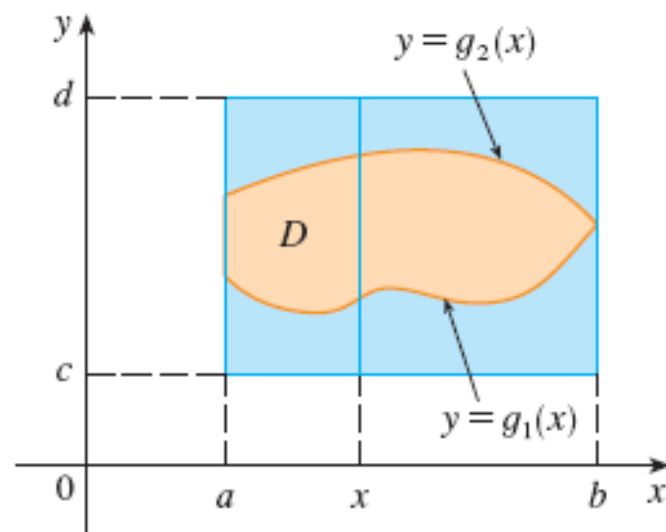


FIGURE 6

3 If f is continuous on a type I region D such that

$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

then

$$\iint_D f(x, y) \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$$

We also consider plane regions of **type II**, which can be expressed as

$$\boxed{4} \quad D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

where h_1 and h_2 are continuous. Two such regions are illustrated in Figure 7.

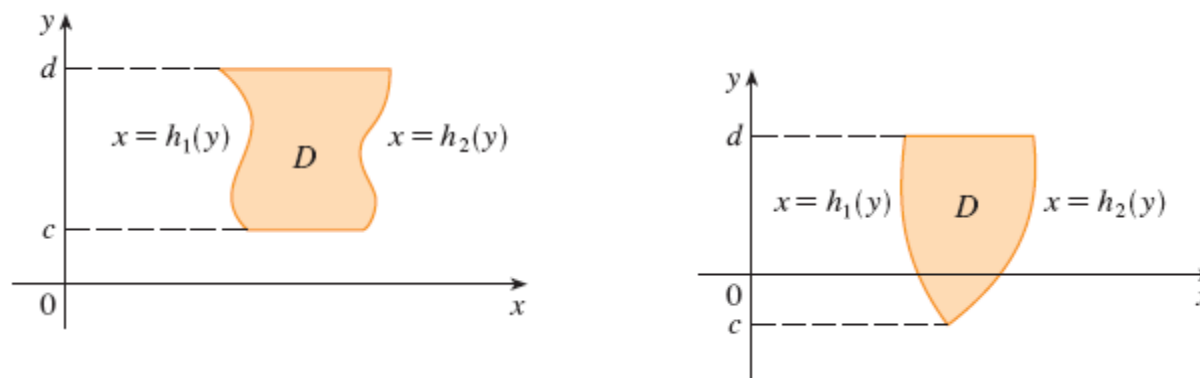


FIGURE 7

Some type II regions

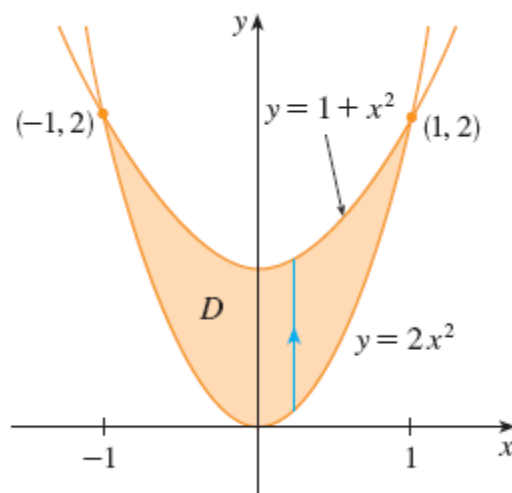
Using the same methods that were used in establishing [3], we can show that

$$\boxed{5} \quad \iint_D f(x, y) \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy$$

where D is a type II region given by Equation 4.

V EXAMPLE 1 Evaluate $\iint_D (x + 2y) \, dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.

$\frac{32}{15}$



EXAMPLE 2 Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region D in the xy -plane bounded by the line $y = 2x$ and the parabola $y = x^2$.

$\frac{216}{35}$

Figure 11 shows the solid whose volume is calculated in Example 2. It lies above the xy -plane, below the paraboloid $z = x^2 + y^2$, and between the plane $y = 2x$ and the parabolic cylinder $y = x^2$.

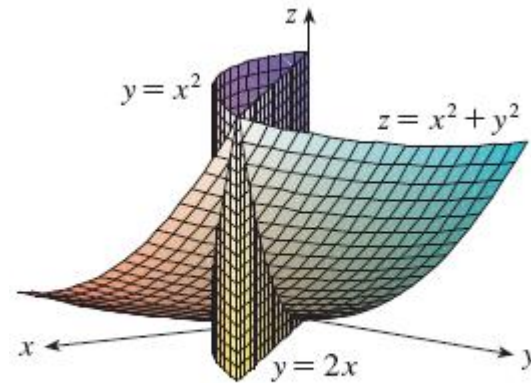


FIGURE 11

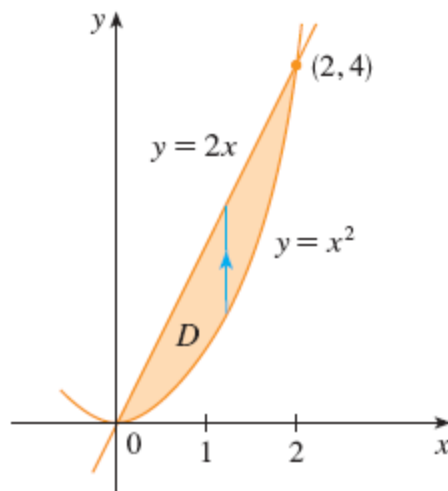


FIGURE 9
 D as a type I region

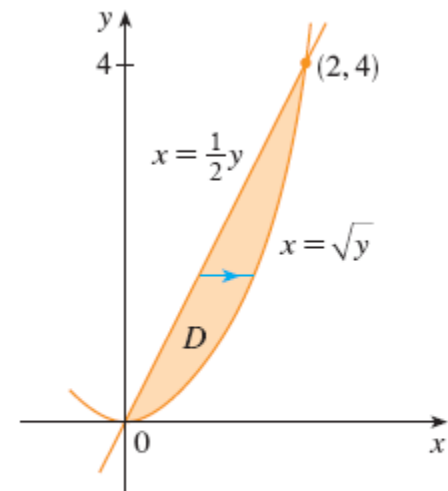


FIGURE 10
 D as a type II region

EXAMPLE 4 Find the volume of the tetrahedron bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$. $\frac{1}{3}$

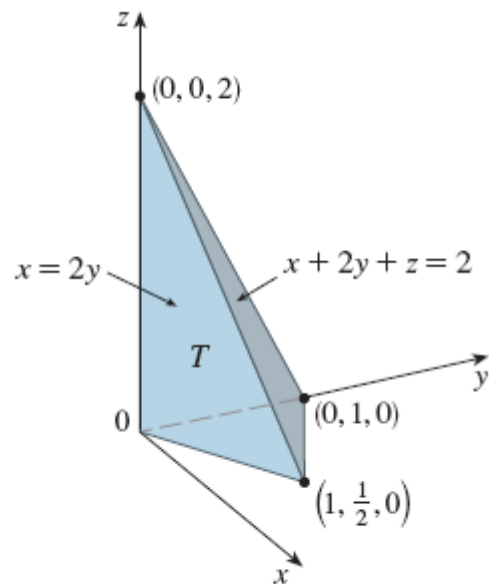


FIGURE 13

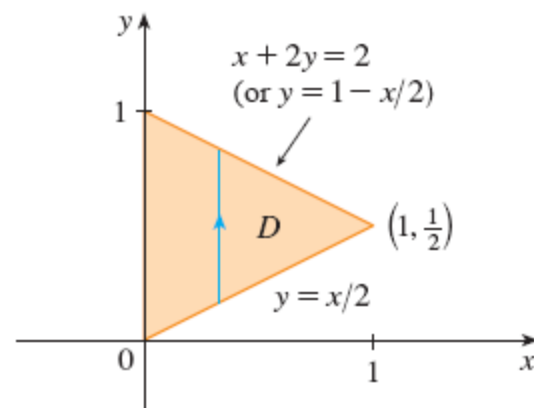


FIGURE 14

Properties of Double Integrals

$$\boxed{6} \quad \iint_D [f(x, y) + g(x, y)] dA = \iint_D f(x, y) dA + \iint_D g(x, y) dA$$

$$\boxed{7} \quad \iint_D c f(x, y) dA = c \iint_D f(x, y) dA$$

If $f(x, y) \geq g(x, y)$ for all (x, y) in D , then

$$\boxed{8} \quad \iint_D f(x, y) dA \geq \iint_D g(x, y) dA$$

The next property of double integrals is similar to the property of single integrals given by the equation $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$.

If $D = D_1 \cup D_2$, where D_1 and D_2 don't overlap except perhaps on their boundaries (see Figure 17), then

9

$$\iint_D f(x, y) dA = \iint_{D_1} f(x, y) dA + \iint_{D_2} f(x, y) dA$$

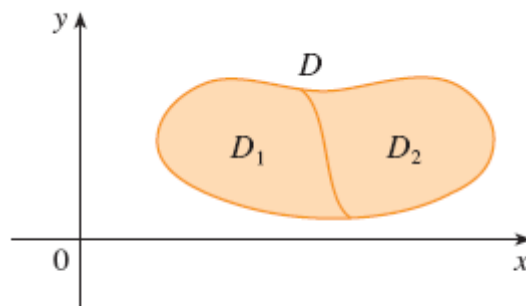


FIGURE 17

Property 9 can be used to evaluate double integrals over regions D that are neither type I nor type II but can be expressed as a union of regions of type I or type II. Figure 18 illustrates this procedure. (See Exercises 55 and 56.)

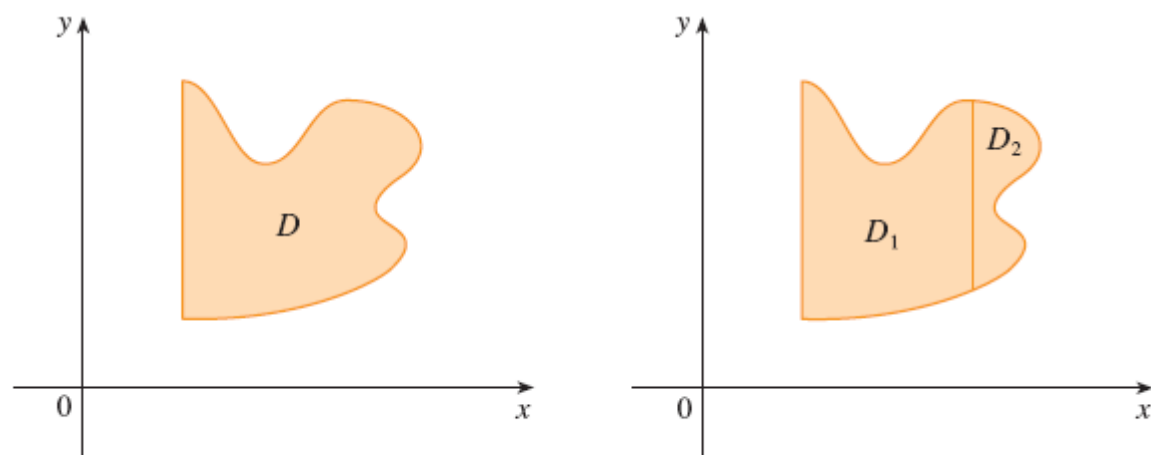


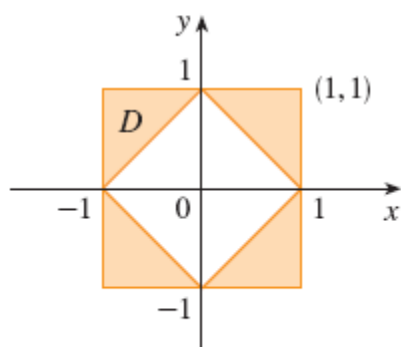
FIGURE 18

(a) D is neither type I nor type II.

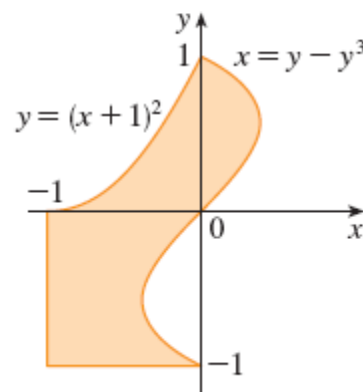
(b) $D = D_1 \cup D_2$, D_1 is type I, D_2 is type II.

55–56 Express D as a union of regions of type I or type II and evaluate the integral.

55. $\iint_D x^2 dA$



56. $\iint_D y dA$



55. 1, 56. -2/15

The next property of integrals says that if we integrate the constant function $f(x, y) = 1$ over a region D , we get the area of D :

10

$$\iint_D 1 \, dA = A(D)$$

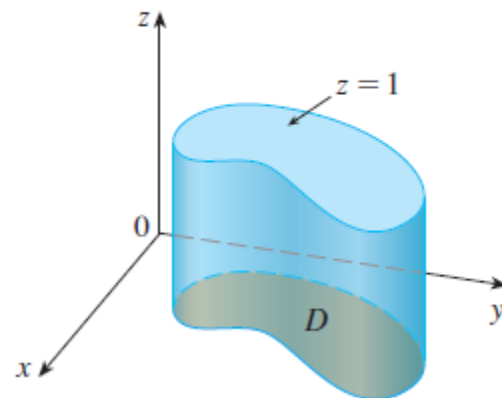


FIGURE 19

Cylinder with base D and height 1

Finally, we can combine Properties 7, 8, and 10 to prove the following property.

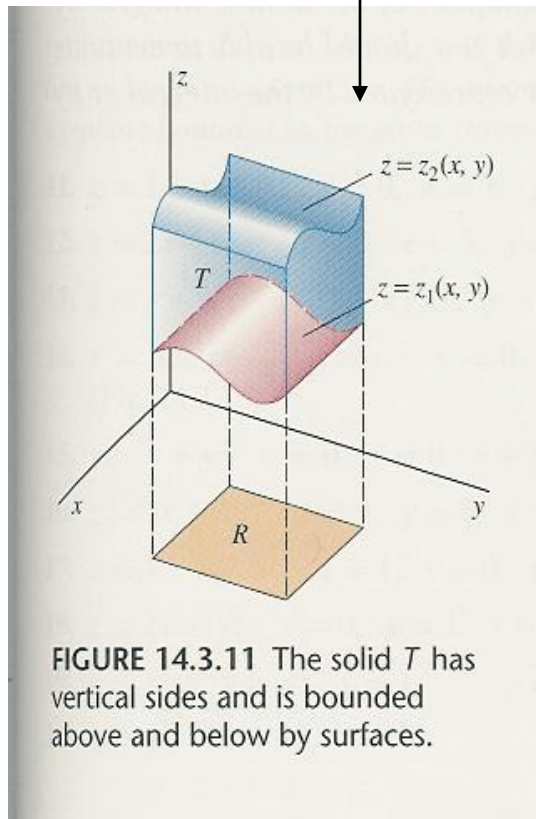
11 If $m \leq f(x, y) \leq M$ for all (x, y) in D , then

$$mA(D) \leq \iint_D f(x, y) \, dA \leq MA(D)$$

Volume Between Two Surfaces

Suppose now that the solid region T lies above the plane region R , as before, but *between* the surfaces $z = z_1(x, y)$ and $z = z_2(x, y)$, where $z_1(x, y) \leq z_2(x, y)$ for all (x, y) in R (Fig. 14.3.11). Then we get the volume V of T by subtracting the volume below $z = z_1(x, y)$ from the volume below $z = z_2(x, y)$, so

$$V = \iint_R [z_2(x, y) - z_1(x, y)] dA. \quad (5)$$



EXAMPLE 4 Find the volume V of the solid T bounded by the planes $z = 6$ and $z = 2y$ and by the parabolic cylinders $y = x^2$ and $y = 2 - x^2$. This solid is sketched in Fig. 14.3.12.

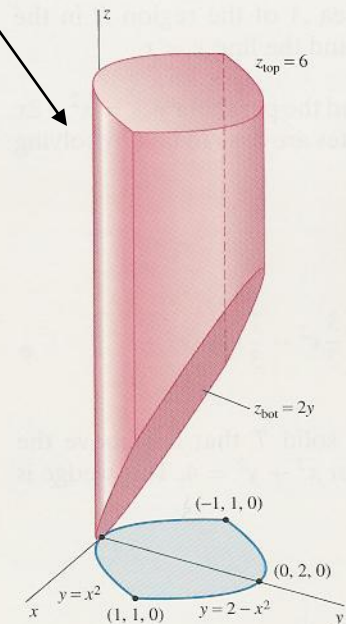
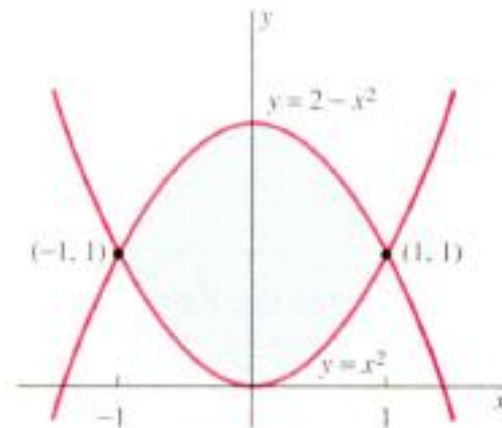
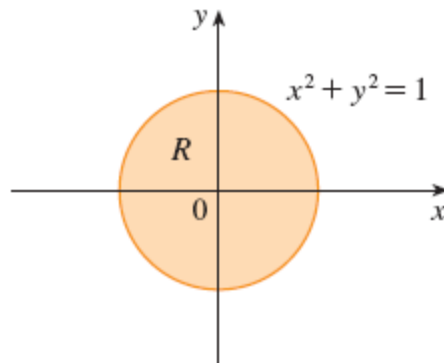
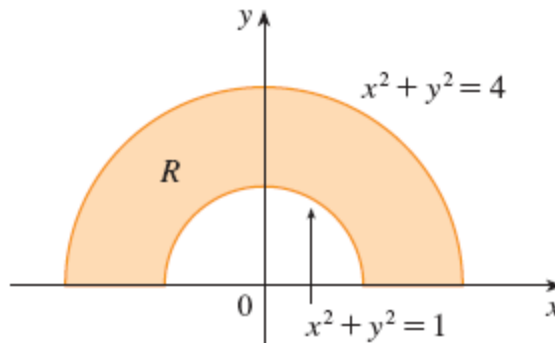


FIGURE 14.3.12 The solid T of Example 4.

Double integrals in polar coordinates



(a) $R = \{(r, \theta) \mid 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$



(b) $R = \{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$

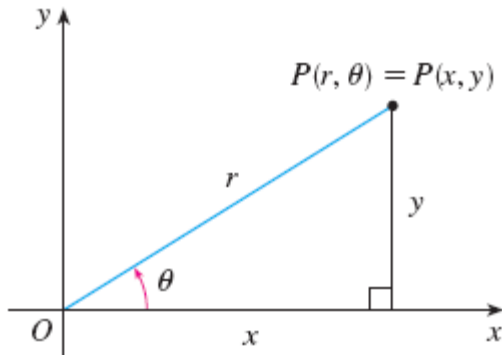


FIGURE 2

$$r^2 = x^2 + y^2 \quad x = r \cos \theta \quad y = r \sin \theta$$

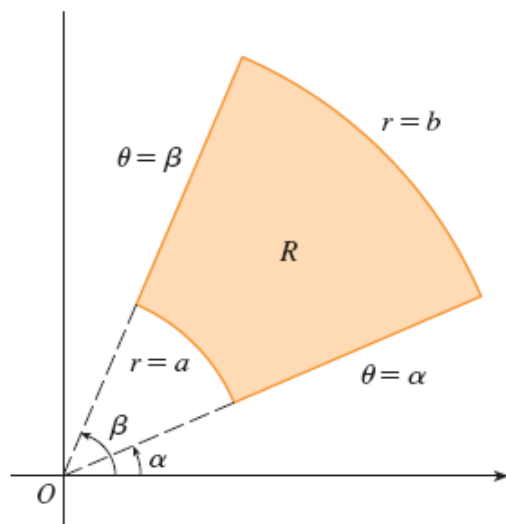


FIGURE 3 Polar rectangle

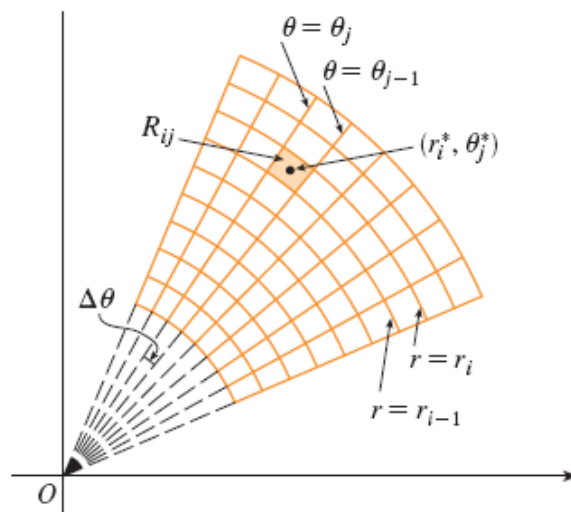


FIGURE 4 Dividing R into polar subrectangles

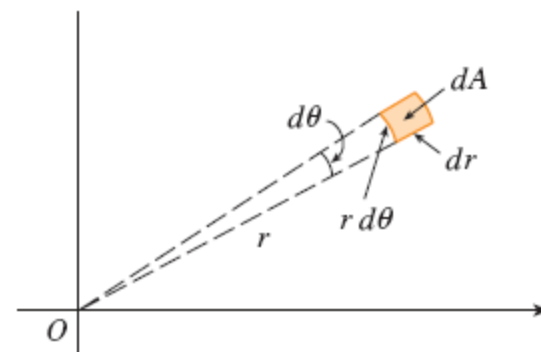


FIGURE 5

$$\boxed{1} \quad \sum_{i=1}^m \sum_{j=1}^n f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) \Delta A_i = \sum_{i=1}^m \sum_{j=1}^n f(r_i^* \cos \theta_j^*, r_i^* \sin \theta_j^*) r_i^* \Delta r \Delta \theta$$

2 Change to Polar Coordinates in a Double Integral If f is continuous on a polar rectangle R given by $0 \leq a \leq r \leq b$, $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$, then

$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$$

V EXAMPLE 2 Find the volume of the solid bounded by the plane $z = 0$ and the paraboloid $z = 1 - x^2 - y^2$. $\frac{\pi}{2}$

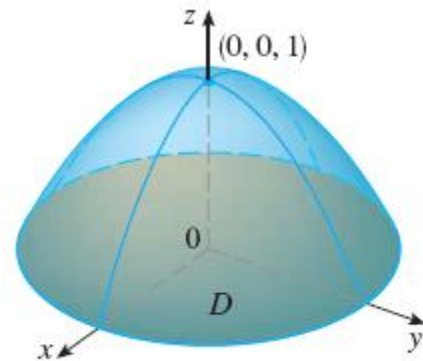


FIGURE 6

More Complicated Region

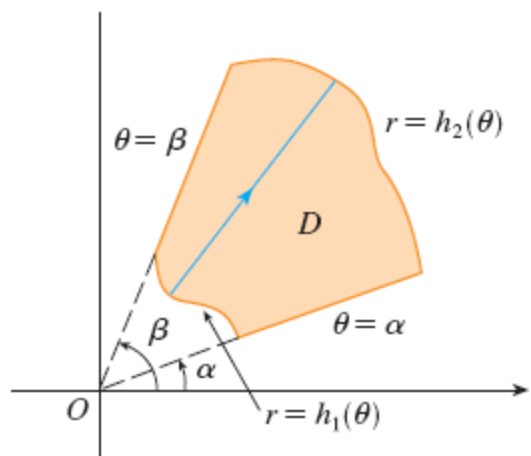


FIGURE 7

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$

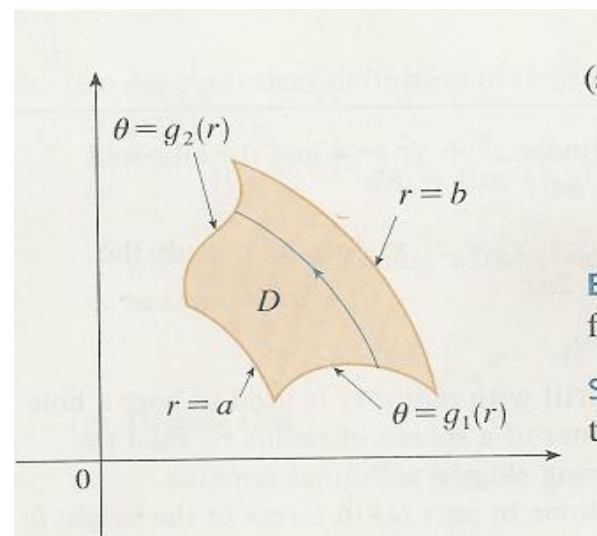


FIGURE 7

Type I polar region,
 $D = \{(r, \theta) \mid a \leq r \leq b, g_1(r) \leq \theta \leq g_2(r)\}$

3 If f is continuous on a polar region of the form

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$

then

$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta$$

$$(4) \quad \iint_D f(x, y) dA = \int_a^b \int_{g_1(r)}^{g_2(r)} f(r \cos \theta, r \sin \theta) r d\theta dr$$

V EXAMPLE 4 Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy -plane, and inside the cylinder $x^2 + y^2 = 2x$.

$$\frac{3\pi}{2}$$

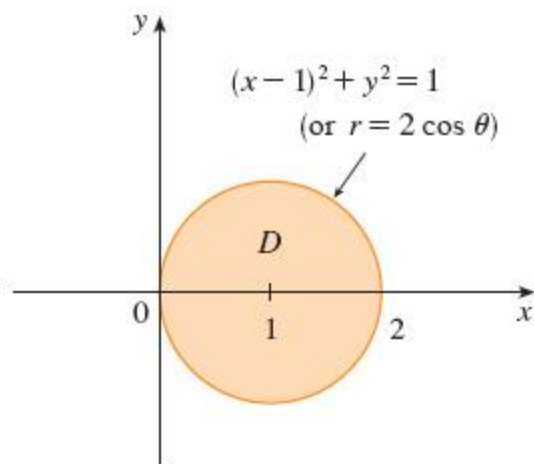


FIGURE 9

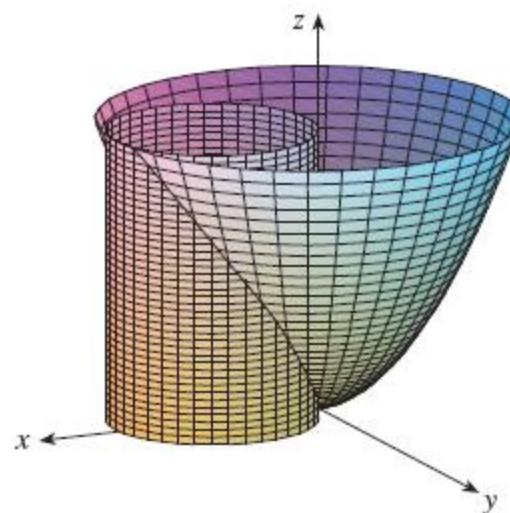


FIGURE 10

Application of Double integrals

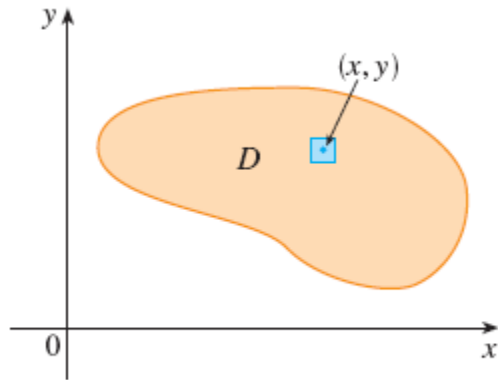


FIGURE 1

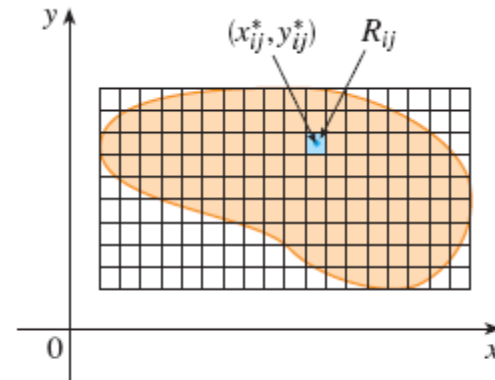


FIGURE 2

Formula for computing mass

Consider a lamina with density function $\rho(x, y)$ that occupies a region D.

1

$$m = \lim_{k, l \rightarrow \infty} \sum_{i=1}^k \sum_{j=1}^l \rho(x_{ij}^*, y_{ij}^*) \Delta A = \iint_D \rho(x, y) dA$$

Moments and centers of mass

Consider a lamina with density function $\rho(x,y)$ that occupies a region D .
Then the moment of the entire lamina about x-axis is

3

$$M_x = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n y_{ij}^* \rho(x_{ij}^*, y_{ij}^*) \Delta A = \iint_D y \rho(x, y) dA$$

Similarly, the moment about the y-axis is

4

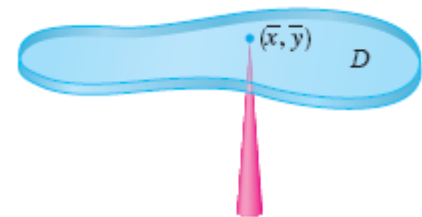
$$M_y = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n x_{ij}^* \rho(x_{ij}^*, y_{ij}^*) \Delta A = \iint_D x \rho(x, y) dA$$

5 The coordinates $(\bar{x}, \bar{y})$ of the center of mass of a lamina occupying the region D and having density function $\rho(x, y)$ are

$$\bar{x} = \frac{M_y}{m} = \frac{1}{m} \iint_D x \rho(x, y) dA \quad \bar{y} = \frac{M_x}{m} = \frac{1}{m} \iint_D y \rho(x, y) dA$$

where the mass m is given by

$$m = \iint_D \rho(x, y) dA$$



V EXAMPLE 2 Find the mass and center of mass of a triangular lamina with vertices $(0, 0)$, $(1, 0)$, and $(0, 2)$ if the density function is $\rho(x, y) = 1 + 3x + y$.

$$m = \frac{8}{3} \quad \bar{x} = \frac{3}{8} \quad \bar{y} = \frac{11}{16}$$

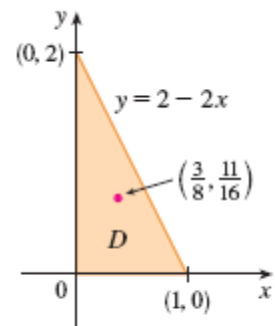
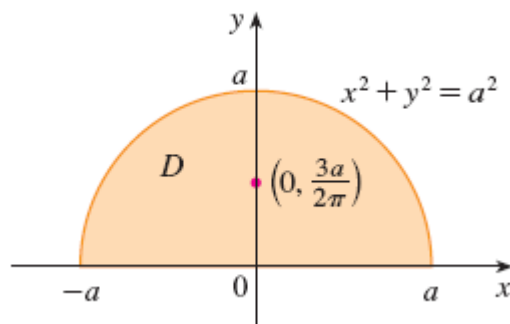


FIGURE 5

V EXAMPLE 3 The density at any point on a semicircular lamina is proportional to the distance from the center of the circle. Find the center of mass of the lamina.



Moments of inertia

The **moment of inertia** (also called the **second moment**) of a particle of mass m about an axis is defined to be mr^2 , where r is the distance from the particle to the axis. We extend this concept to a lamina with density function $\rho(x, y)$ and occupying a region D by proceeding as we did for ordinary moments. We divide D into small rectangles, approximate the moment of inertia of each subrectangle about the x -axis, and take the limit of the sum as the number of subrectangles becomes large. The result is the **moment of inertia** of the lamina **about the x -axis**:

$$6 \quad I_x = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n (y_{ij}^*)^2 \rho(x_{ij}^*, y_{ij}^*) \Delta A = \iint_D y^2 \rho(x, y) dA$$

Similarly, the **moment of inertia about the y -axis** is

$$7 \quad I_y = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n (x_{ij}^*)^2 \rho(x_{ij}^*, y_{ij}^*) \Delta A = \iint_D x^2 \rho(x, y) dA$$

It is also of interest to consider the **moment of inertia about the origin**, also called the **polar moment of inertia**:

$$8 \quad I_0 = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n [(x_{ij}^*)^2 + (y_{ij}^*)^2] \rho(x_{ij}^*, y_{ij}^*) \Delta A = \iint_D (x^2 + y^2) \rho(x, y) dA$$